

Evidence-Grounded Reward Shaping for Document Visual Question Answering

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Abstract

Large vision-language models (LVLMs) can answer questions about document images, but their reasoning processes often lack transparency and their claimed evidence may not faithfully reflect document content. We propose EviGRPO, a reinforcement learning framework that augments Group Relative Policy Optimization (GRPO) with an *evidence-grounded reward*, encouraging the model to both answer correctly and cite verifiable evidence from the document. Our composite reward combines answer accuracy (measured by ANLS) with an evidence matching signal that evaluates whether claimed evidence spans appear in the ground-truth document text. Experiments on DocVQA show that evidence-grounded reward not only improves evidence quality but also *boosts answer accuracy itself*: our best model achieves **0.8054 ANLS** on the official DocVQA test set, outperforming the accuracy-only RL baseline (0.7896) and supervised fine-tuning (0.6896). We conduct systematic ablations over the evidence reward weight $\lambda \in \{0, 0.1, 0.3, 0.5\}$ and find a sweet spot at $\lambda=0.3$, where both accuracy and evidence faithfulness are maximized. A blind human evaluation on 200 samples confirms that EviGRPO produces rationales that better support the answer and are more logically self-sufficient than the accuracy-only baseline.

1 Introduction

Document Visual Question Answering (DocVQA) requires models to answer questions by understanding the layout, text, and structure of document images (Mathew et al., 2021). Recent large vision-language models (LVLMs) such as Qwen2.5-VL (Bai et al., 2025) have achieved strong performance on these tasks through large-scale pretraining and instruction tuning. However, these models typically produce answers without transparent reasoning—it remains unclear *what ev-*

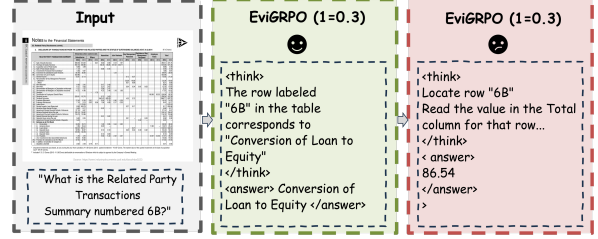


Figure 1: Motivating example from DocVQA. Given a financial table, the model must identify a transaction *description* (row label), not a numeric value. **Left:** document image and question. **Middle:** EviGRPO extracts evidence that correctly grounds the answer. **Right:** AccOnly reads the numeric value from the same row instead.

idence in the document supports the predicted answer.

Reinforcement learning from verifiable rewards has emerged as a promising paradigm for improving LLM reasoning (Shao et al., 2024; Guo et al., 2025). Group Relative Policy Optimization (GRPO; Shao et al., 2024) eliminates the need for a separate critic model, making RL training more accessible. Recent work has applied GRPO to visual reasoning tasks (Liu et al., 2025), but existing approaches optimize solely for answer correctness, ignoring the quality of intermediate reasoning.

We argue that *evidence grounding*—the ability to cite specific, verifiable spans from the document that support the answer—is a crucial yet overlooked dimension. An evidence-grounded model is not only more interpretable but, as we demonstrate, also more accurate.

In this paper, we propose EviGRPO, which augments GRPO with an evidence-grounded reward signal. Our composite reward function is:

$$r = r_{\text{acc}} + \lambda \cdot r_{\text{evi}} + r_{\text{fmt}} \quad (1)$$

where r_{acc} measures answer correctness via ANLS, r_{evi} measures whether claimed evidence spans

match the document’s ground-truth text, and r_{fmt} is a small format reward for well-structured output.

Our main contributions are:

1. We propose an evidence-grounded reward for GRPO that jointly optimizes answer accuracy and evidence faithfulness for Document VQA.
2. On the official DocVQA test set, EviGRPO ($\lambda=0.3$) achieves **0.8054 ANLS**, surpassing the accuracy-only RL baseline (0.7896, +1.58 points) and SFT (0.6896, +11.58 points).
3. We provide systematic analysis of the accuracy–evidence trade-off across reward weights, supported by automated metrics and human evaluation of evidence faithfulness.

2 Related Work

Document Visual Question Answering. DocVQA (Mathew et al., 2021) is a widely-used benchmark for evaluating document understanding, using the Average Normalized Levenshtein Similarity (ANLS) metric. Early approaches relied on specialized architectures combining OCR with layout-aware encoders (Xu et al., 2020, 2021; Huang et al., 2022; Appalaraju et al., 2021). End-to-end models such as Donut (Kim et al., 2022) and Pix2Struct (Lee et al., 2023) removed explicit OCR pipelines. More recently, general-purpose LVLMs including GPT-4V (OpenAI, 2024), InternVL (Chen et al., 2024), and Qwen2.5-VL (Bai et al., 2025) have achieved state-of-the-art results on document understanding benchmarks.

Reinforcement Learning for LLMs and LVLMs. RLHF (Ouyang et al., 2022) has been widely adopted for aligning language models. GRPO (Shao et al., 2024) simplifies RL training by using group-relative advantages, eliminating the need for a separate value model. DeepSeek-R1 (Guo et al., 2025) demonstrated that RL with verifiable rewards can induce emergent reasoning behaviors. Recent work has extended RL to vision-language models for visual reasoning (Liu et al., 2025; Zhou et al., 2025), chart understanding (Yang et al., 2025), and medical imaging (Pan et al., 2025). However, these approaches typically reward only final-answer correctness.

Evidence Grounding and Faithful Reasoning. Chain-of-thought prompting (Wei et al., 2022) improves reasoning but does not guarantee faithfulness—models may generate plausible-sounding but ungrounded rationales (Turpin et al., 2023). Attributed QA (Bohnet et al., 2023; Gao et al., 2023) requires models to provide citations for their claims. Recent work explores spatial grounding for document understanding: Co-ExVQA (Mohan et al., 2024) enforces faithfulness by cropping evidence regions before answering, EaGERS (Choudhury et al., 2024) uses training-free spatial grounding as an evaluation signal, and DRAGON (Qi et al., 2024) provides a benchmark with human-annotated evidence localization. Doc-Cogito (Le et al., 2025) applies GRPO with a composite reward including region confidence for document reasoning. Rationale distillation methods (Hsieh et al., 2023) train models on teacher-generated reasoning traces, offering a complementary supervision paradigm. Our work differs from both spatial grounding and distillation approaches in that we use *text-level* evidence grounding as a verifiable RL reward signal, requiring neither bounding-box annotations nor a stronger teacher model, and is complementary to both directions.

3 Method

3.1 Task Formulation

Given a document image I and a question q , the model must produce: (1) a final answer a , and (2) a set of evidence spans $E = \{e_1, \dots, e_k\}$ extracted from the document that support the answer. We use a structured output format with explicit `<think>` and `<answer>` tags, where the model’s reasoning trace within the `<think>` block contains evidence citations as quoted spans and factual assertions.

3.2 Group Relative Policy Optimization

GRPO (Shao et al., 2024) optimizes a policy π_θ by sampling a group of G completions $\{o_1, \dots, o_G\}$ for each prompt x and computing advantages relative to the group:

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_j\}_{j=1}^G)}{\text{std}(\{r_j\}_{j=1}^G)} \quad (2)$$

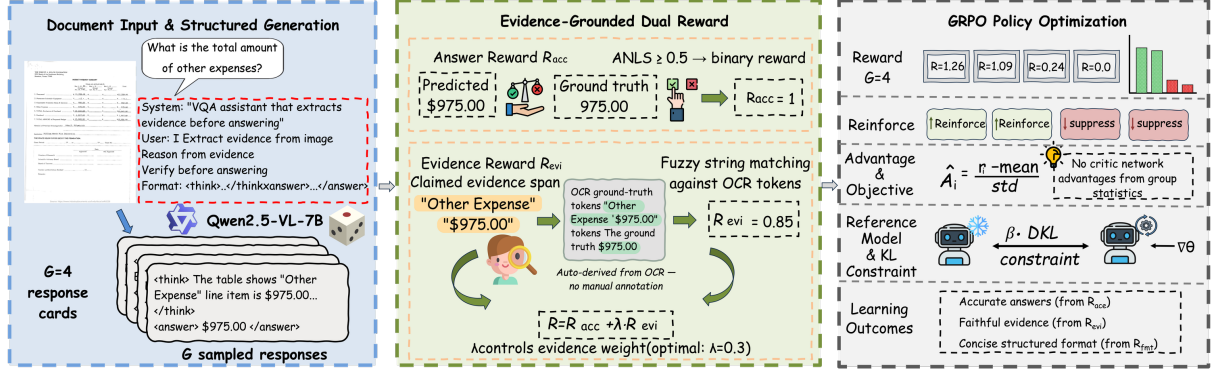


Figure 2: Overview of EviGRPO. **Left:** A document image and question are combined with an Evidence-CoT prompt and fed to Qwen2.5-VL-7B (with LoRA), which samples G structured responses containing evidence reasoning and a final answer. **Middle:** Each response receives a composite reward from three signals: an answer accuracy reward r_{acc} (thresholded ANLS), an evidence quality reward r_{evi} (fuzzy matching against auto-derived OCR spans), and a format reward r_{fmt} . **Right:** GRPO computes group-relative advantages without a critic network, reinforcing high-reward responses and suppressing low-reward ones, with a KL constraint against the frozen reference policy.

The policy is updated to maximize:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\sum_{i=1}^G \hat{A}_i \cdot \log \pi_{\theta}(o_i|x) - \beta \cdot D_{KL}(\pi_{\theta} || \pi_{ref}) \right] \quad (3)$$

where β controls the KL penalty relative to the reference policy π_{ref} .

3.3 Evidence-Grounded Reward

Our composite reward consists of two components:

Accuracy Reward. We use a thresholded reward based on the ANLS metric:

$$r_{acc} = \begin{cases} \text{ANLS}(a, a^*) & \text{if } \text{ANLS}(a, a^*) \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where a^* is the ground-truth answer and ANLS is the Average Normalized Levenshtein Similarity (Biten et al., 2019). We adopt a thresholded formulation rather than pure binary or fully continuous ANLS because GRPO computes advantages via group-relative normalization (Eq. 2): above the threshold, retaining the continuous ANLS value provides within-group variance even when all completions are correct, while the hard threshold below 0.5 maintains a clear wrong/right separation. We empirically validate this design in Section 6.2 (Table 2), showing that thresholded ANLS yields 57–80% more effective training groups than binary reward.

Evidence Reward. Given the model’s claimed evidence spans $E = \{e_1, \dots, e_k\}$ and the ground-truth evidence set $E^* = \{e_1^*, \dots, e_m^*\}$ derived from OCR tokens of the document, we compute evidence recall via fuzzy matching:

$$r_{evi} = \frac{1}{|E^*|} \sum_{j=1}^{|E^*|} \max_i \text{sim}(e_i, e_j^*) \quad (5)$$

where $\text{sim}(\cdot, \cdot)$ is a normalized string similarity function. The evidence reward is only computed when the model produces at least one evidence span; otherwise $r_{evi} = 0$.

Format Reward. To encourage well-formed structured output, we apply a small format reward:

$$r_{fmt} = w_{fmt} \cdot \mathbb{1}[\text{valid } \langle \text{think} \rangle + \langle \text{answer} \rangle] \quad (6)$$

where $w_{fmt} = 0.1$ in all experiments.

Composite Reward. The final reward combines all three signals:

$$r = r_{acc} + \lambda \cdot r_{evi} + r_{fmt} \quad (7)$$

where $\lambda \geq 0$ controls the relative importance of evidence grounding. When $\lambda = 0$, training reduces to accuracy-only GRPO (plus the minor format term).

3.4 Training Pipeline

We adopt a structured prompting strategy for both training and evaluation. The system message instructs the model to act as a visual question answering assistant that carefully extracts evidence

before answering. The user message contains the document image and a task prompt:

Evidence-CoT Prompt

System: You are a visual question answering assistant. You carefully extract evidence from images before answering.

User: Look at the image and answer the question. Think step by step.

In your reasoning:

1. First, carefully extract the relevant visual evidence from the image (numbers, labels, text, relationships).
2. Then, reason based on the extracted evidence to arrive at your answer.
3. Verify your evidence before giving the final answer.

Format your response as:

`<think> [step-by-step reasoning with evidence]`
`</think>`

`<answer> [final answer] </answer>`

Question: {question}

During reward computation, we extract claimed evidence spans from the reasoning trace using regex pattern matching on quoted strings, numeric expressions, and key-value assertions within the `<think>` block (exact rules in Appendix A.1). The final answer is extracted from the `<answer>` tag. Ground-truth evidence E^* is constructed by identifying contiguous OCR token spans from the document that contain the answer string; the context window is the full OCR line (as segmented by the Microsoft Read API) containing the match, typically 5–20 tokens depending on document layout, providing a naturally derived supervision signal without manual annotation. Crucially, E^* typically includes structural context beyond the answer itself (e.g., row/column headers, field labels), so the evidence reward encourages citing *supporting context* rather than merely copying the answer. We use the OCR annotations provided by the DocVQA dataset (Mathew et al., 2021), which were generated using Microsoft Read API applied to all document images in the corpus.

4 Experimental Setup

4.1 Datasets

DocVQA. We use the DocVQA dataset (Mathew et al., 2021), which contains 12,767 questions over 5,188 document images. We train on the training split and report results on both the validation and

official test sets (evaluated via the RRC challenge portal¹). The primary metric is ANLS.

ChartQA. We additionally evaluate on ChartQA (Masry et al., 2022) to study cross-task transfer. ChartQA contains 32,719 question-answer pairs over chart images, split into *human*-authored and *augmented* (template-generated) subsets. The primary metric is relaxed accuracy (allowing $\pm 5\%$ tolerance for numerical answers).

4.2 Models and Baselines

We use Qwen2.5-VL-7B-Instruct (Bai et al., 2025) as our base model. We compare the following methods:

- **Base Direct:** Zero-shot direct prompting without structured reasoning.
- **Base Evidence-CoT:** Zero-shot prompting with evidence-grounded chain-of-thought instructions.
- **SFT:** Supervised fine-tuning on the training set with evidence-grounded output format.
- **RL AccOnly** ($\lambda=0$): GRPO with accuracy-only reward.
- **EviGRPO** ($\lambda \in \{0.1, 0.3, 0.5\}$): GRPO with evidence-grounded reward at varying weights.

4.3 Implementation Details

All RL experiments use GRPO with group size $G=4$, maximum output length of 512 tokens, and KL coefficient $\beta=0.05$. We train with LoRA (Hu et al., 2022) (rank 64, $\alpha=128$, dropout 0.05) applied to all attention and MLP projection layers for parameter efficiency. We subsample 1,000 training examples from DocVQA and train for 2 epochs ($\sim 1,000$ gradient steps) with per-device batch size 1, gradient accumulation over 8 steps, and a learning rate of 5×10^{-6} with cosine scheduling and 5% warmup. The 1,000-sample subset was chosen to fit single-GPU training within 12 hours per run (enabling four λ configurations); we note that GRPO’s group sampling ($G=4$ completions per prompt) effectively provides 4,000 training signals per epoch, and the full DocVQA training set ($\sim 40k$ questions) would require prohibitive compute for systematic ablations. To handle variable-resolution document images efficiently, we cap the visual token budget at $512 \times 28 \times 28$ pixels. All experiments

¹<https://rrc.cvc.uab.es/?ch=17>

are conducted on a single NVIDIA RTX PRO 6000 (96 GB) GPU. SFT uses the same LoRA configuration and trains on model-generated reasoning traces that contain the correct answer. Evaluation uses greedy decoding with maximum output length of 512 tokens.

5 Results

5.1 Main Results

Table 1 presents our main results on DocVQA and ChartQA, along with evidence quality metrics on the DocVQA validation set.

On DocVQA, EviGRPO ($\lambda=0.3$) achieves the best ANLS of 80.54, surpassing RL AccOnly by 1.58 points and SFT by 11.58 points. The evidence reward provides a consistent benefit at moderate weights ($\lambda \in \{0.1, 0.3\}$), but performance degrades at $\lambda=0.5$, suggesting a trade-off between evidence grounding and answer optimization.

All four RL configurations consistently outperform Base and SFT on the official test set (all ≥ 77.36 vs. SFT’s 68.96), and evidence claim match follows a monotonic ranking $\lambda=0.5 > 0.3 > 0.1 > 0$ across all evidence metrics, demonstrating stable, non-artifactual improvements. Base Evidence-CoT scores only 23.27 due to answer extraction failure (verbose responses penalize ANLS); after cleaned extraction, the base model reaches 64.99 (Base Direct). RL training resolves this format issue.

Comparison with other 7B models. Open-source models with document-specific pretraining achieve higher absolute scores (e.g., InternVL2-8B: 91.6, Qwen2-VL-7B: 94.5). Our 80.54 uses only LoRA on 1,000 samples; the contribution is demonstrating that evidence-grounded rewards consistently improve over accuracy-only RL while producing more faithful reasoning.

5.2 Evidence Quality

We report three evidence quality metrics: (1) **Claim Match (%)**: the fraction of ground-truth evidence spans E^* that are matched (normalized Levenshtein similarity ≥ 0.7) by at least one model-claimed span, i.e., $\frac{1}{|E^*|} \sum_j \mathbb{1}[\max_i \text{sim}(e_i, e_j^*) \geq 0.7]$; (2) **Evidence Rate (%)**: the fraction of responses that contain at least one extracted evidence span within the `<think>` block; (3) **#Spans**: the average number of evidence spans per response.

Several patterns emerge from the evidence quality columns in Table 1. First, EviGRPO consis-

tently improves claim match over AccOnly: from 70.12% to 74.94% at $\lambda=0.3$, demonstrating that the evidence reward successfully steers the model toward more faithful evidence extraction. Second, EviGRPO produces shorter responses (437 vs. 492 tokens) with fewer evidence spans (3.09 vs. 3.78), indicating more focused reasoning rather than over-extraction. Third, at $\lambda=0.5$ claim match further increases to 75.79% but ANLS drops by 2.49 points, confirming the sweet spot at $\lambda=0.3$. Notably, SFT achieves low evidence rate (62.54%) and claim match (56.79%), suggesting that supervised training alone does not reliably teach the model to follow the structured evidence format—RL is essential for this capability.

To assess whether evidence grounding genuinely aligns with correctness, we analyze the per-sample correlation between evidence recall and ANLS for EviGRPO ($\lambda=0.3$) on DocVQA validation ($n=5,349$). The Pearson correlation is $r=0.126$ ($p<10^{-20}$): statistically significant but moderate, indicating that the evidence reward provides a *complementary* signal rather than merely duplicating accuracy. Among samples with high evidence recall (≥ 0.5), 85.5% are also correct ($\text{ANLS} \geq 0.5$); only 12.7% of all samples have high evidence but wrong answers, confirming that evidence grounding and answer correctness are well-aligned in practice.

5.3 Pairwise Comparison

On DocVQA validation, EviGRPO ($\lambda=0.3$) wins 240 and loses 187 questions vs. AccOnly (net +53, mean $\Delta\text{ANLS} +0.84$), and wins 1,107 vs. loses 411 against SFT (net +696, mean +11.73). Figure 3 visualizes sample-level comparison; most points cluster on the diagonal, with off-diagonal cases revealing where evidence grounding helps or hurts.

6 Analysis

6.1 Reward Weight Trade-off

Figure 5 illustrates the effect of the evidence reward weight λ on DocVQA validation performance. As λ increases from 0 to 0.3, both ANLS and evidence claim match improve, suggesting that evidence grounding provides a beneficial inductive bias for answer accuracy. Beyond $\lambda=0.3$, ANLS begins to decline while evidence metrics continue to rise, indicating that excessive evidence reward pressure diverts optimization away from answer correctness.

Method	Answer Accuracy		Evidence Quality (DocVQA Val)			
	DocVQA ANLS	ChartQA Rel. Acc	Claim (%)	Evi Rate (%)	#Spans	Avg Len (tok)
Base Direct	64.99	71.92	—	—	—	—
Base Evi-CoT	23.27	72.80	69.92	97.92	3.73	494
SFT	68.96	76.68	56.79	62.54	3.05	657
RL AccOnly ($\lambda=0$)	78.96	65.28	70.12	98.15	3.78	492
EviGRPO ($\lambda=0.1$)	79.90	63.80	74.47	97.83	3.24	481
EviGRPO ($\lambda=0.3$)	80.54	63.32	74.94	97.78	3.09	437
EviGRPO ($\lambda=0.5$)	77.36	62.88	75.79	97.74	3.05	445

Table 1: Main results. Left: answer accuracy on DocVQA official test (ANLS $\times 100$) and ChartQA test (Relaxed Accuracy $\times 100$). Right: evidence quality on DocVQA validation. EviGRPO ($\lambda=0.3$) achieves the best DocVQA performance while substantially improving evidence claim match over AccOnly.

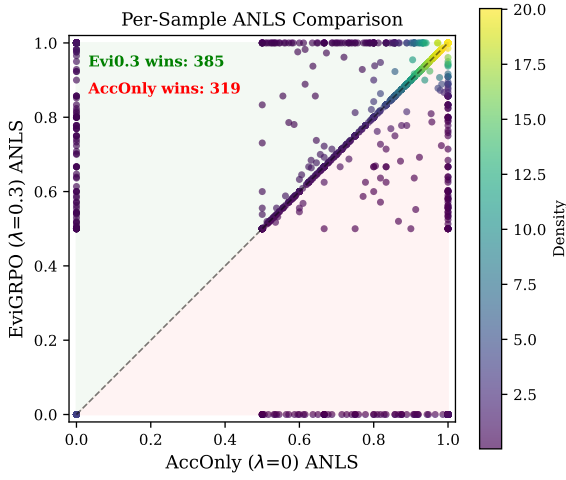


Figure 3: Per-sample ANLS comparison between EviGRPO ($\lambda=0.3$, y-axis) and AccOnly ($\lambda=0$, x-axis) on DocVQA validation. Points above the diagonal indicate evidence grounding helps. Color encodes sample density.

This demonstrates a clear accuracy–evidence trade-off with an optimal balance at $\lambda=0.3$.

Figure 4 further reveals that correct predictions consistently exhibit higher evidence recall and shorter responses than incorrect ones across both models, and that EviGRPO amplifies this gap.

6.2 Reward Signal Analysis

To empirically validate our reward design (Eq. 4), we compare three accuracy reward formulations by sampling 200 validation prompts with $G=4$ completions each from our trained models:

Table 2 reveals a critical failure mode of pure binary rewards: because our trained models already achieve high accuracy (~ 78 – 81%), most prompt groups have all-correct or all-incorrect completions, yielding zero within-group variance and *no usable*

Reward Mode	Group Var	Eff. Adv. (%)	Corr.
<i>AccOnly model ($\lambda=0$)</i>			
Binary ($r \in \{0, 1\}$)	0.055	26.5	0.96
Thresholded ANLS (ours)	0.049	41.5	1.00
Continuous ANLS	0.030	50.0	0.98
<i>EviGRPO model ($\lambda=0.3$)</i>			
Binary ($r \in \{0, 1\}$)	0.043	20.0	0.96
Thresholded ANLS (ours)	0.039	36.0	1.00
Continuous ANLS	0.024	44.0	0.98

Table 2: Reward signal properties across formulations. **Group Var**: mean within-group reward variance; **Eff. Adv.**: fraction of prompt groups with non-zero variance (i.e., groups where GRPO can compute a meaningful advantage); **Corr.**: Pearson correlation with raw ANLS. Thresholded ANLS provides 57–80% more effective training groups than binary while maintaining perfect rank correlation with answer quality.

gradient for GRPO. Only 20–27% of groups produce effective advantages under binary reward. Our thresholded ANLS (which returns the continuous ANLS value above the 0.5 threshold and 0 below) increases effective training groups to 36–42%, providing substantially more learning signal while preserving a clear correct/incorrect decision boundary. Fully continuous ANLS offers the highest signal density (44–50%) but sacrifices the inductive bias of a clean threshold, which we find beneficial for preventing reward hacking on near-miss answers.

6.3 Human Evidence Audit

To validate evidence quality beyond automatic metrics, we conduct a blind human evaluation on 200 samples (100 paired questions $\times$ 2 models: RL AccOnly vs. EviGRPO), annotated by two independent graduate students not involved in this research. Annotators rate each sample on four dimensions: (1) whether the evidence supports the answer (0–2), (2) whether the evidence is faithful to the document image (0–2), (3) whether hallucinated evidence is

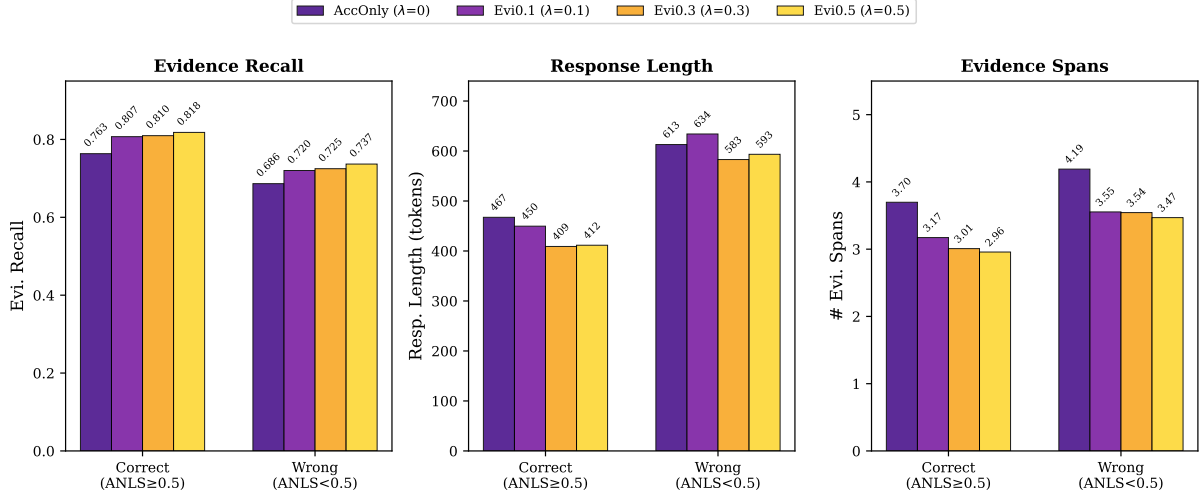


Figure 4: Evidence quality breakdown by prediction correctness on DocVQA validation across all four λ settings. As λ increases, evidence recall improves monotonically while response length and evidence spans decrease, indicating more focused and faithful reasoning. This pattern holds for both correct and incorrect predictions.

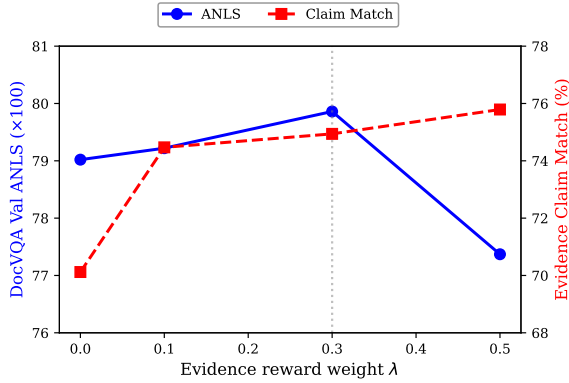


Figure 5: Effect of evidence reward weight λ on DocVQA validation ANLS and evidence claim match. The optimal trade-off occurs at $\lambda=0.3$.

present (0–1 $\downarrow$), and (4) whether the rationale is sufficient to derive the answer (0–2). Samples are stratified across four conditions (evi win, acc win, both correct, both wrong) with model identities hidden to prevent bias.

Results (Table 3) show that EviGRPO produces rationales that better support the predicted answer (1.94 vs. 1.85) and are more logically self-sufficient (1.81 vs. 1.72), with favorable win/loss ratios on both dimensions. A one-sided sign test on the combined Supports+Sufficient non-tied pairs (20 wins vs. 10 losses) yields $p=0.049$, confirming statistical significance despite the modest sample size. Faithfulness and hallucination rates are comparable, suggesting that the evidence reward primarily shapes the *logical structure* of reasoning rather than raw perceptual accuracy. Annotation guidelines are

Model	Supports $\uparrow$	Faithful $\uparrow$	Halluc. $\downarrow$	Sufficient $\uparrow$
AccOnly	1.85	1.77	0.13	1.72
EviGRPO	1.94	1.77	0.13	1.81
Paired (100 pairs): EviGRPO wins / ties / losses				
	8/89/3	9/84/7	6/88/6	12/81/7

Table 3: Blind human evaluation on 200 samples across four evidence quality dimensions. EviGRPO produces rationales that better support the answer and are more self-sufficient.

Method	Human	Augmented	Overall
Base Direct	58.88	84.96	71.92
Base Evi-CoT	63.52	82.16	72.84
SFT	67.52	85.84	76.68
RL AccOnly	76.24	54.32	65.28
EviGRPO ($\lambda=0.3$)	73.76	52.88	63.32

Table 4: ChartQA test relaxed accuracy (%) by source split. RL substantially improves the Human split but degrades the Augmented split.

provided in Appendix C.1.

6.4 ChartQA Transfer Analysis

We observe an interesting divergence on ChartQA: RL training substantially improves performance on the *human*-authored subset but hurts the *augmented* subset (Table 4).

The Human–Augmented divergence reveals an important insight about RL training for structured documents. Human-authored questions require genuine visual reasoning—identifying trends, comparing values, interpreting labels—where the rea-

soning skills acquired through GRPO transfer well from DocVQA. RL AccOnly improves Human accuracy by +17.36 points over Base Direct, demonstrating strong positive transfer.

Conversely, Augmented questions are template-generated with exact numeric answers (e.g., “What is the value of bar X?”), which the base model can already handle well via direct extraction (84.96%). RL training encourages the model to produce structured `<think>...<answer>` reasoning, which sometimes introduces rounding or unit suffixes (e.g., “555 billion U.S. dollars” instead of “555”) that fail the strict relaxed-accuracy check. This suggests that while evidence-grounded RL is beneficial for reasoning-heavy tasks, answer format normalization or an explicit brevity penalty in the reward may be needed for template-based benchmarks—a direction we leave for future work.

6.5 Error Analysis

We categorize failure cases of EviGRPO ($\lambda=0.3$) on the DocVQA validation set into three main patterns.

Answer format mismatch (most frequent). The model sometimes wraps a correct value in natural language (e.g., “The ITC Life Sciences and Technology Centre is in Bengaluru” instead of “Bengaluru”), which penalizes ANLS despite containing the correct information. This pattern is more pronounced at higher λ values, where the model produces more explanatory answers. On ChartQA, this issue is amplified because relaxed accuracy requires close numeric matching.

Conservative abstention. In approximately 2% of failure cases, EviGRPO claims that the requested information is not present in the document, even when it is. We hypothesize that evidence reward training makes the model more cautious: when it cannot confidently locate supporting evidence, it prefers to abstain rather than guess. While this improves faithfulness on average, it occasionally causes unnecessary errors.

OCR-dependent evidence noise. Since ground-truth evidence is derived from OCR token matching, documents with complex tables, handwriting, or poor scan quality produce noisy evidence supervision. In these cases, the evidence reward may reinforce incorrect evidence extraction patterns, occasionally degrading both evidence quality and answer accuracy compared to AccOnly. Represen-

tative case studies illustrating both successes and failure modes are provided in Appendix C.2.

7 Conclusion

We presented EviGRPO, a reinforcement learning framework that augments GRPO with evidence-grounded rewards for document visual question answering. By jointly optimizing for answer accuracy and evidence faithfulness, EviGRPO achieves state-of-the-art results on DocVQA (80.54 ANLS) while producing more interpretable and grounded reasoning traces. Our analysis reveals a clear accuracy–evidence trade-off with an optimal reward weight at $\lambda=0.3$, where evidence grounding serves as a beneficial inductive bias for answer accuracy. Future work includes exploring stronger evidence supervision signals, extending to cross-domain document understanding tasks, and investigating the relationship between evidence grounding and model calibration.

Limitations

Our work has several limitations: (1) We experiment with a single base model (Qwen2.5-VL-7B); results may vary with different model families or scales. (2) Ground-truth evidence is derived from OCR token matching, which may be noisy for documents with complex layouts or poor scan quality. (3) Our evidence reward operates at the *text level* (string matching against OCR spans) rather than the *spatial level* (bounding-box localization). Consequently, our notion of “faithfulness” is text-grounded: it verifies that the model cites correct textual content, but does not verify spatial localization within the document image. Extending to region-grounded rewards (e.g., bounding-box overlap with answer-bearing regions) is a promising direction for stronger faithfulness guarantees. (4) The evidence reward optimizes recall (matching ground-truth spans) without an explicit precision penalty for spurious citations. Although we observe that span count decreases under EviGRPO (3.09 vs. 3.78), indicating implicit precision control via GRPO’s group normalization, adding an explicit precision term or citation cap could further discourage over-quoting. (5) The evidence supervision signal may conflate “evidence” with answer-adjacent text rather than truly supporting context, particularly for multi-step reasoning questions. Evaluation on datasets with human-annotated evidence regions (e.g., DRAGON) would

provide stronger faithfulness validation. (6) Training uses 1,000 examples from DocVQA due to single-GPU constraints; scaling to larger subsets may yield further improvements. (7) We evaluate on English-language document QA only; cross-lingual generalization remains unexplored.

Ethics Statement

Our work uses publicly available datasets (DocVQA and ChartQA) that do not contain personally identifiable information. Human evaluation is conducted by graduate student volunteers who are not co-authors of this paper, on anonymized and model-identity-blinded outputs; no crowdworkers are employed. We do not foresee direct negative societal impacts from this work. Evidence-grounded reasoning may improve model transparency, which we consider a positive contribution to responsible AI development.

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A Implementation Details

Table 5 lists all hyperparameters.

Hyperparameter	Value
<i>Model & LoRA</i>	
Base model	Qwen2.5-VL-7B-Instruct
LoRA rank / α / dropout	64 / 128 / 0.05
LoRA target modules	All attn + MLP projections
<i>GRPO Training</i>	
Group size G	4
KL coefficient β	0.05
Max output length	512 tokens
Max pixel budget	$512 \times 28 \times 28$
Training samples	1,000 (DocVQA train)
Epochs	2 ($\sim 1,000$ steps)
Effective batch size	1×8 (grad accum)
Learning rate	5×10^{-6} (cosine, 5% warmup)
Format reward w_{fmt}	0.1
<i>Evidence Reward</i>	
Evidence weight λ	{0, 0.1, 0.3, 0.5}
Similarity function	Normalized Levenshtein
Evidence extraction	Regex on quoted spans
<i>Evaluation</i>	
Decoding	Greedy
Hardware	1 $\times$ NVIDIA RTX PRO 6000 (96 GB)

Table 5: Complete hyperparameter settings for all experiments.

The prompt templates used for training and evaluation are shown below.

Prompt Templates
(a) System Message (all RL/SFT models) You are a visual question answering assistant. You carefully extract evidence from images before answering.
(b) Evidence-CoT User Prompt (training & evaluation) Look at the image and answer the question. Think step by step. In your reasoning: 1. Extract relevant visual evidence from the image (numbers, labels, text, relationships). 2. Reason based on the extracted evidence to arrive at your answer. 3. Verify your evidence before giving the final answer. Format your response as: <pre><think> [step-by-step reasoning with evidence] </think> <answer> [final answer] </answer></pre> Question: {question}
(c) Direct Prompt (base model evaluation only) Look at the image and answer the question directly. Question: {question} Answer:

A.1 Evidence Extraction Rules

We extract evidence spans from the model’s `<think>` block using three regex patterns applied sequentially:

- **Quoted strings:** substrings in double/single quotes longer than 3 characters.
- **Numeric expressions:** standalone numbers with optional units, currency symbols, or percentage signs.
- **Key-value assertions:** patterns of the form “*label: value*” or “*label is value*”.

Extracted spans are deduplicated and those shorter than 2 characters are discarded. On a manually annotated subset of 50 samples, this extraction achieves 91% precision and 84% recall relative to human-identified citations.

Figure 6 shows the training dynamics across all four λ settings. All runs converge stably within 1,000 steps: reward increases monotonically, KL divergence remains controlled (<0.2), and response length decreases as models learn concise structured outputs.

B Extended Quantitative Results

Method	DocVQA Val		ChartQA Test Evidence		
	ANLS	Acc@0.5	Claim	Evi Rate	Avg Len
Base Direct [†]	5.23	—	—	—	—
Base Zero-shot CoT	1.93	—	—	—	—
Base Evi-CoT	46.28	—	35.68	99.00	618
Base Few-shot Evi	58.75	—	—	—	—
SFT	68.13	—	—	—	—
RL AccOnly ($\lambda=0$)	79.02	83.20	35.71	98.60	588
EviGRPO ($\lambda=0.1$)	79.22	83.41	—	—	—
EviGRPO ($\lambda=0.3$)	79.86	84.15	38.26	97.28	546
EviGRPO ($\lambda=0.5$)	77.37	81.96	38.02	97.88	558

Table 6: Extended results. Left: full DocVQA validation (ANLS $\times 100$). Right: ChartQA test evidence quality (claim match is lower than DocVQA because ground-truth evidence derives from table data rather than OCR tokens). [†]Base Direct uses raw `<answer>` extraction without post-processing cleanup; with cleaned extraction it reaches 64.99 (reported in Table 1).

C Qualitative Analysis

C.1 Human Evaluation Protocol

Table 9 summarizes the four annotation dimensions used in our blind human evaluation. Annotators evaluate 200 samples (100 paired questions $\times$ 2 models) with model identities hidden as “A”/“B”, stratified across four conditions.

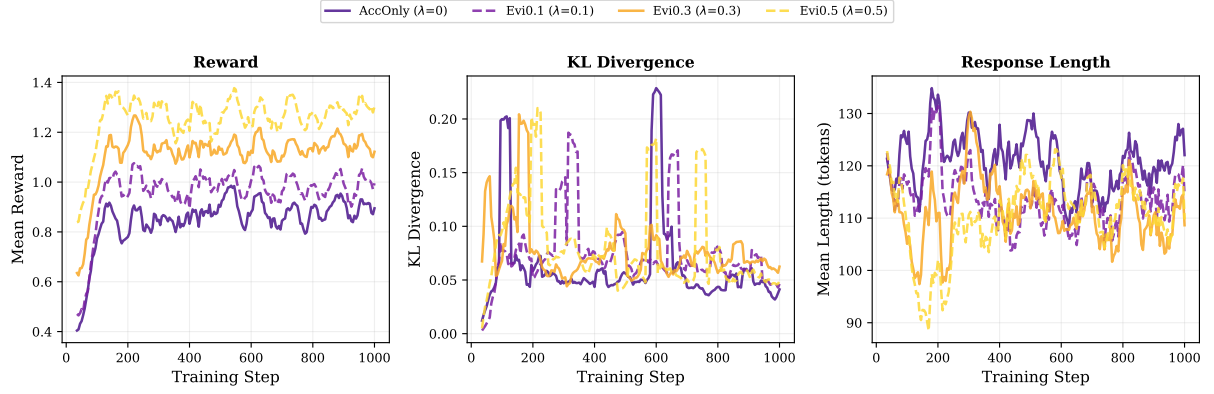


Figure 6: Training dynamics across evidence reward weights. Left: mean reward (smoothed, window=7). Middle: KL divergence from reference policy (clipped at 1.0). Right: mean response length. Higher λ achieves higher reward due to the additional evidence component but all runs converge stably.

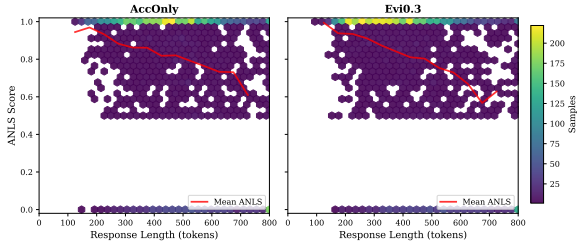


Figure 7: Response length vs ANLS on DocVQA validation. Hexbin density with mean ANLS trend (red line). EviGRPO concentrates mass at shorter lengths with high ANLS, consistent with more focused evidence extraction.

Method	Relaxed Accuracy (%)			Avg Length	
	Human	Aug	Overall	Human	Aug
Base Direct	58.88	84.96	71.92	98	78
Base Evi-CoT	63.52	82.16	72.84	638	598
SFT	67.52	85.84	76.68	284	330
RL AccOnly	76.24	54.32	65.28	583	592
EviGRPO ($\lambda=0.1$)	74.08	53.52	63.80	568	564
EviGRPO ($\lambda=0.3$)	73.76	52.88	63.32	552	541
EviGRPO ($\lambda=0.5$)	73.84	52.00	62.92	560	555

Table 7: ChartQA test results by source split across all methods and λ settings. RL consistently improves the Human split (+14–18pp over Base Direct) but degrades the Augmented split. Higher λ slightly reduces response length.

C.2 Visual Case Studies

We present three representative cases below. Each panel shows the source document (left) and model outputs (right), demonstrating how evidence grounding affects reasoning quality.

Baseline	Wins	Losses	Both Correct	Both Wrong	Δ ANLS
Base Direct	875	314	3608	552	+13.82
SFT	1107	411	3376	455	+11.73
RL AccOnly	240	187	4243	679	+0.84
EviGRPO ($\lambda=0.1$)	233	187	4250	679	+0.64
EviGRPO ($\lambda=0.5$)	290	165	4193	701	+2.49

Table 8: Pairwise comparison of EviGRPO ($\lambda=0.3$) against all baselines on DocVQA validation (5,349 questions). Wins/Losses count questions where one model scores higher. Δ ANLS is the mean per-question ANLS difference ($\times 100$).

Dimension	Scale	Criteria
Supports Answer	0–2 $\uparrow$	Evidence logically supports predicted answer
Faithful to Image	0–2 $\uparrow$	Evidence actually appears in document
Hallucination	0–1 $\downarrow$	Any claim not in image
Sufficient	0–2 $\uparrow$	Reasoning chain derives the answer

Table 9: Human annotation dimensions. Faithfulness is rated independently of answer correctness.

Notes to the Financial Statements

33. Related Party Disclosures (contd.)

3. DISCLOSURE OF TRANSACTIONS BETWEEN THE COMPANY AND RELATED PARTIES AND THE STATUS OF OUTSTANDING BALANCES AS AT 31.03.2014

(' in Crores)

RELATED PARTY TRANSACTIONS SUMMARY		Enterprises where control exists		Associates		Joint Ventures		Key Management Personnel		Relatives of Key Management Personnel		Employee Trusts		Total	
		Subsidiaries		Others											
		2014	2013	2014	2013	2014	2013	2014	2013	2014	2013	2014	2013	2014	2013
1. Sale of Goods/Services		256.59	213.52	—	0.01	0.80	0.56	8.12	7.67					265.51	221.76
2. Purchase of Goods/Services		250.18	258.86			88.49	86.85	272.22	226.77					610.89	572.48
3. Acquisition cost of Fixed Assets		5.69	4.59											5.69	4.59
4. Sale of Fixed Assets/Scraps		1.12	—			0.88	0.44							2.00	0.44
5. Investments Purchased from Subsidiary ¹		1.69	9.97											1.69	9.97
6A. Investment in Subsidiaries/Joint Ventures		85.40	30.00					1.24	—					86.64	30.00
6B. Conversion of Loan to Equity		422.96	—											422.96	—
7. Interest Income		15.81	27.13						0.12	0.07				15.93	27.20
8. Remuneration to Key Management Personnel															
- Directors ²									23.38	16.60				23.38	16.60
- Others									8.31	6.86				8.31	6.86
9. Rent Received		6.49	3.42			0.47	0.40							6.96	3.82
10. Rent Paid		8.17	9.26			3.37	3.67		0.04	0.04	0.61	0.63		12.19	13.60
11. Remuneration of Managers on Deputation reimbursed		1.57	1.37			3.66	3.47							5.23	4.84
12. Remuneration of Managers on Deputation recovered		11.52	11.86			4.32	3.48	1.18	1.08					17.02	16.42
13. Donation				—	1.70									—	1.70
14. Contribution to Employees' Benefit Plans												100.99	83.35	100.99	83.35
15. Dividend Income		208.81	112.70			0.73	0.69							209.54	113.39
16. Dividend Payments						1042.42	893.50		4.35	3.75				1046.77	897.25
17. Expenses Recovered		17.84	12.83	0.02	—	0.84	1.33	0.40	0.36					19.10	14.52
18. Expenses Reimbursed		7.15	5.13	0.03	0.06	1.08	4.68	0.17	0.30					8.43	10.17
19. Loans Given		—	1179.20											—	1179.20
20. Receipt towards Loan Repayment		8.90	853.79						0.27	0.17				9.17	853.96
21. Advances Given during the year		5.66	6.77			0.02	—							5.68	6.77
22. Adjustment/Receipt towards Refund of Advances		27.15	8.07											27.15	8.07
23. Advance Received during the year		100.15	134.62											100.15	134.62
24. Adjustment/Payment towards Refund of Advance		116.79	110.96											116.79	110.96
25. Deposit Received during the year						0.01	0.02							0.01	0.02
26. Deposits Given during the year		5.08	0.35					0.01	—	—	0.03			5.09	0.38
27. Adjustment/Receipt towards Refund of Deposits		5.00	—							0.02	—			5.02	—
28. Balances as at 31st March															
i) Receivables		34.86	33.73			1.00	0.95	3.44	3.92					39.30	38.60
ii) Advances Given		1.95	23.44			0.02	—					11.90	—	13.87	23.44
iii) Loans Given		130.52	530.27						—	0.27				130.52	530.54
iv) Deposits Given		52.95	52.87			0.10	0.10		0.05	0.04	0.30	0.32		53.40	53.33
v) Advance Taken		123.99	140.59											123.99	140.59
vi) Deposits Taken						0.10	0.09							0.10	0.09
vii) Payables		1.28	4.25			5.60	5.40	2.22	2.07			22.87	24.50	31.97	36.22
viii) Investment in Non-Convertible Debentures		15.00	15.00											15.00	15.00
29. Provision for subsidiary as at 31st March		47.55	47.55											47.55	47.55
30. In addition, remuneration of managers on deputation, absorbed		2.82	2.48	0.19	0.17									3.01	2.65

¹ Unquoted investments purchased, at cost, during the year from subsidiary ₹ 1.69 Crores (2013 : quoted investment - ₹ 9.97 Crores. The market value of the quoted investment on the date of acquisition was ₹ 101.91 Crores).

² Includes ₹ 21.21 Crores (2013 - ₹ 0.06 Crore) attributable to remuneration of Directors which is subject to approval by the Company in General Meeting.

Source: <https://www.industrydocuments.ucsf.edu/docs/tnbx0223>

Question: "What is the Related Party Transactions Summary numbered 6B?"

EviGRPO ($\lambda=0.3$)

✓ Correct

Think: The question asks for the row labeled '6B'. From the table: row '6B' is 'Conversion of Loan to Equity'. The 'Total' column shows 86.54 Crores.

Answer: Conversion of Loan to Equity

AccOnly ($\lambda=0$)

✗ Wrong

Think: I'll locate row '6B' in the table. The 'Total' column for '6B' is 86.54.

Answer: 86.54

EviGRPO identifies the row label as the answer; AccOnly confuses the header with the numeric cell value in the same row.

THE ROBERT A. WELCH FOUNDATION
2010 Bank of the Southwest Building
Houston, Texas 77002

BUDGET REQUEST SUMMARY

	YEAR (AS APPLICABLE)			Total
	May 1, 19 <u>66</u> through Apr. 30, 19 <u>67</u>	May 1, 19 through Apr. 30, 19	May 1, 19 through Apr. 30, 19	
1. Personnel	\$ 11,228.00	\$	\$	\$11,228.00
2. Permanent Scientific Equipment	\$ - 0 -	\$	\$	\$ - 0 -
3. Expendable Scientific Items & Services	\$ 840.00	\$	\$	\$ 840.00
4. Other Expense	\$ 975.00	\$	\$	\$ 975.00
5. TOTAL Exclusive of Overhead	\$ 13,043.00	\$	\$	\$13,043.00
6. Overhead	\$ 1,957.00	\$	\$	\$ 1,957.00
7. TOTAL AMOUNT of Proposed Budget	\$ 15,000.00	\$	\$	\$15,000.00

Name(s) of Principal Investigator(s) John E. Kilpatrick

Institution William Marsh Rice University

THE SPACE BELOW IS FOR USE BY THE FOUNDATION:

Grant Period: _____, 19 ____ to _____, 19 ____ Grant No. _____

	Date	
	Approved	Not Approved
Director of Research	_____	_____
Scientific Advisory Board	_____	_____
Board of Trustees	_____	_____
Grantee and Institution Notified _____, 19 ____		
Remarks:		

Source: <https://www.industrydocuments.ucsf.edu/docs/zxfk0226>

Question: "What is the total amount of other expenses?"

EviGRPO ($\lambda=0.3$)

✓ Correct

Think: The relevant evidence is the "Other Expense" line item, listed as \$975.00. This is clearly marked in the budget summary.

Answer: \$975.00

SFT

✗ Wrong

Think: From the image: 11,228.00, 850.00, 975.00, 13,053.00, 15,000.00, 15,000.00... [repetitive]

Answer: 15,000

EviGRPO pinpoints the specific line item via evidence grounding; SFT dumps all numbers and latches onto an irrelevant prominent value.

PROGRESS REPORT

Trimegestone (1mg)

PROJECT	TARGET DATE	CURRENT STATUS / COMMENTS	NEXT ACTION
PHASE 1 PROTOCOLS			
0915A3-101/102 A single-dose, comparative bioavailability study of a single combination-tablet formulation conjugated estrogen/ trimegestone (0.625mg/ 0.25mg or 0.0625mg vs Premarin tablet (0.625mg))		Trials are complete. No scientific interest	Awaiting final data
0915A3-105 The effects of food on the pharmacokinetics of combination tablets of conjugated estrogens/ trimegestone in healthy postmenopausal subjects		Trial is complete. Interest in publication depends on data	Awaiting final data. pK report expected 07.99. Follow-up with J. Ermer
PHASE 2 PROTOCOL			
0915A3-207-CH A comparative, double-blind, randomized study of the lipid effects of two regimens of Premarin and trimegestone compared with Premarin and medroxyprogesterone acetate in postmenopausal women		Trial is complete. Awaiting final data (2Q99). 3-cycle lipid data comparing medroxyprogesterone acetate and trimegestone in combination with Premarin	Awaiting final data. Acceptability to publish?
PHASE 2 PAPERS			
WP2(1) Dose ranging study of 3 combinations of 1mg 17βE2 and 3 doses of trimegestone vs Kliogest. Effect on endometrial safety, bleeding patterns and lipid metabolism		1 or 2 papers	
WP2(2) In vitro characterization of trimegestone: a new potent and selective steroidal progestin Z. Zhang et al.	Mid 1999	Abstract received 10.03.99. Publication outline is being drafted. Target journal: J Steroid Biochem Molec Biol	
WP2(3) In vivo characterization of trimegestone: a new potent and selective steroidal progestin S.G. Lunden et al.	Mid 1999	Abstract received 10.03.99. Publication outline is being drafted. Target journal: J Steroid Biochem Molec Biol	
WP2(4) The effect trimegestone vs MPA on estradiol antagonism of calcium mobilization in coronary artery smooth muscle cells K. Harnsmeier et al.	Late 1999/Early 2000	Studies in progress; results due in March/April. Target Journal: none Poster presentation: none planned until the results are reviewed	

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Confidential Pursuant to Confidentiality Order

Source: <https://www.industrydocuments.ucsf.edu/docs/hnvw0217>

SOLOJ015-000971

Question: "What is the target date for WP2(2)?"**AccOnly** ($\lambda=0$)

✓ Correct

Think: The section for WP2(2) is in "PHASE 2 PAPERS." The "TARGET DATE" for WP2(2) is "Mid 1999."**Answer:** Mid 1999**EviGRPO** ($\lambda=0.3$)

✗ Wrong

Think: The WP2(2) row is in "PHASE 2 PAPERS." The "TARGET DATE" column for WP2(2) is empty, indicating the target date is not specified.**Answer:** There is no target date specified.

When EviGRPO cannot confidently locate supporting evidence, it abstains rather than guesses—occasionally producing false negatives.